Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms a correct equation by applying conservation of energy with KE and PE − substituting the values given	AO3.4	M1	KE gained = PE lost $\frac{1}{2}mv^2 = mgh$ $\frac{1}{2}(75)v^2 = 75g(25)$ $v^2 = 50g$ Or $v^2 = 490$ $v = 22$ m s^{-1} (2sf)
	Solves the equation correctly to obtain the value of v − must be rounded correctly to 2 significant figures	AO3.2a	A1	
(b)	Forms expressions for EPE and PE	AO3.1b	M1	PE lost = EPE gained $\frac{\lambda x^2}{2l} = mgh$ $\frac{3200x^2}{2(25)} = 75(9.8)(25+x)$ $64x^2 - 735x - 18375 = 0$ $x = 23.6\ldots$ $23.6 + 25 = 48.6$ m 48.6 m < 50 m Dominic does not get wet
	Obtains fully correct expressions	AO1.1b	A1	
	Solves a three-term quadratic equation or substitutes correct distances for a total length of 50 m into both expressions.	AO1.1a	M1	
	Obtains the correct maximum extension of the cord or obtains the correct values for the two energies	AO1.1b	A1	
	Compares ‘their’ results and concludes that Dominic does not get wet	AO3.2a	E1F	
(c)	Comments that Dominic has size and considers the effect this might have on the distance fallen	AO3.2b	E1	Dominic has size so distance descended would be greater than 48.6 m
	States clearly whether Dominic does or does not get wet, justifying their conclusion	AO3.5a	E1F	He might get wet if he was taller than 1.4 m
				Total 9 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an energy equation between PE and EPE	AO3.3	M1	$0.5 \times 2 \times 10 = \frac{\lambda}{2 \times 0.2} \times 0.4^2$
	Obtains correct value for λ or k	AO1.1b	A1	
	Using Hooke’s Law to find	AO3.4	M1	

	extension at equilibrium			$\lambda = \frac{10}{0.4}$ $= 25$ Nm^{-1} $2 \times 10 = \frac{25}{0.2}e$ $e = \frac{2 \times 10 \times 0.2}{25}$ $= 0.16$ $2 \times 10 \times 0.26 = \frac{1}{2} \times 2v^2 + \frac{25 \times 0.16^2}{2 \times 0.4}$ $v^2 = 3.6$ $v = 1.897\ldots$ $= 2.0$ m s^{-1}(to 1sf)
	Obtains correct extension at equilibrium using 'their' λ or k	AO1.1b	A1F	
	Forms equation using conservation of energy.	AO3.4	M1	
	Obtains the correct speed. (Only accept $V = 2$ m s^{-1} or $V = 2.0$ m s^{-1})	AO1.1b	A1	
(b)	States appropriate refinement.	AO3.5c	E1	Could include air resistance.
				Total 7 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates correct EPE.	AO1.1b	B1	$\text{EPE} = \frac{65 \times 0.7^2}{2 \times 1.3} = 12.25\text{J}$ Friction 2 × 9.8 × cos(30°) × 0.6 = 10.184 N EPE = $F_r(2 - d)$ + $mg(2 - d)\sin(30°)$ + $\frac{65(d-1.3)^2}{2 \times 1.3}$ 12.25 = 10.184(2 – d) + 9.8(2 – d) + 25(d – 1.3)^2 $25d^2 - 84.984d + 69.968 = 0$
	Finds correct value for friction.	AO3.1b	B1	
	Forms work-energy equation, with at least two correct terms.	AO3.3	M1	
	Obtains work-energy equations with correct terms and correct signs.	AO1.1b	A1	
	Solves 'their' energy equation to obtain two solutions. (PI)	AO1.1a	M1	
	Rejects d = 2 with a reason.	AO2.4	A1	
	Completes rigorous argument to obtain 1.4.	AO2.1	R1	

AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips. All energy terms must be included and two solutions stated for the quadratic equation.			d = 1.39936 or 2 d = 2 represents the starting position so is not a solution given that the particle moves d = 1.4 m
(b) Calculates the component of weight parallel to the plane.	AO3.1b	M1	Component of weight down plane = 2×9.8sin(30°) = 9.6
Compares 'their' (sum of) friction (and tension) with the component of weight parallel to the plane.	AO2.4	R1	Limiting value of friction = 10.184 > 9.6 (and there is tension too.)
Determines that particle does not slide back down plane.	AO2.2a	R1	∴ Particle does not slide back down the plane.
			Total 10 marks

Q4.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,dx$

SC1 $\int_0^e \frac{\lambda e}{l}\,de$

M1

$= \left[\frac{\lambda x^2}{2l}\right]_0^e$

SC1 $\int \frac{\lambda x}{l}\,dx$ *with no limits*

A1

$= \frac{\lambda e^2}{2l}$

A1

3

(b) (i) Using $T = \frac{\lambda x}{l}$:

$5g = \frac{392x}{1.6}$

$x = \frac{5g \times 1.6}{392}$

= 0.2

M1

Extension is 0.2 m

A1

2

(ii) When extension is 0.6 m, EPE = $\frac{\lambda x^2}{2l}$

B1 for 0.6

B1

= $\frac{392 \times (0.6)^2}{2 \times 1.6}$

M1

= 44.1 J

A1

3

(iii) Let y metres be distance particle is above A.

C of energy, when particle has speed 0.8 m s^{-1}, gives

$5 \times g \times y + \frac{392 \times (0.6 - y)^2}{2 \times 1.6} + \frac{1}{2} \times 5 \times 0.8^2$

M1 4 terms, 2 correct

M1A1 4 terms, 3 correct

M1A1

= $\frac{392 \times (0.6)^2}{2 \times 1.6}$

M1A2 4 terms correct

Ft answer to (b)(ii)

A1F

$49y + 122.5(0.6 - y)^2 + 1.6 = 122.5 \times 0.6^2$

$49y - 147y + 122.5y^2 + 1.6 = 0$

$122.5y^2 - 98y + 1.6 = 0$

$y = \frac{98 \pm \sqrt{98^2 - 4 \times 122.5 \times 1.6}}{2 \times 122.5}$

$y = \frac{98 \pm 93.9148}{245}$

= 0.016674 and 0.7833

if x used instead of 0.6 – y, A1 here for x = 0.5833...

A1

Speed first becomes 0.8 when y = 0.0167

E1

5

[13]

Q5.

(a) (i) Using $T = \frac{\lambda x}{l}$

Tension in string is $\frac{60 \times 2.5}{3}$
= 50 N

B1

Frictional force on A [using $F = \mu R$]
is 0.4 × 8 × g
= 31.36 N

B1

which is less than tension in string
Thus particle A moves towards the hole

B1

3

(ii) Gravitational force on B is 3g = 29.4
which is less than tension in string

B1

Thus particle B moves towards the hole

B1

2

(b) EPE = $\frac{\lambda x^2}{2l}$

$= \frac{60 \times (2.5)^2}{2 \times 3}$

M1

= 62.5 J

A1

2

(c) Let x be the distance B has moved upwards

Work done by friction [on A] is 31.36 × 0.46

M1

= 14.4256
= 14.43 J

A1

When B is at rest, extension is 2.04 – x

EPE = $\frac{\lambda x^2}{2l}$

$= \frac{60 \times (2.04 - x)^2}{2 \times 3}$

= $10(2.04 - x)^2$ J

B1

C of Energy, when particle B is at rest, gives
$3 \times g \times x + 10(2.04 - x)^2 + 14.4256$

M1A1

= 62.5

A1

$10x^2 - 11.4x - 6.4584 = 0$

x = 1.555 and – 0.415

Or $10x^2 - 11.4x - 6.454 = 0$

Particle B is first at rest when it has moved upwards 1.56 m

Accept 1.55

A1

7

[14]

Q6.

(a) using EPE = $\frac{\lambda x^2}{2l}$,

M1

EPE = $\frac{32\times2.2^2}{2\times0.8}$

B1 for 2.2

B1

= 96.8 J

A1

3

(b) by C of Energy, when next at rest, EPE (initial) = work done against friction + EPE (when at rest)

M1A1 for work done by friction or 5F

M1A1

96.8 = F × 5 + $\frac{32\times1.2^2}{2\times0.8}$

M1 3 terms; A1 all correct

M1A1

5F = 96.8 – 28.8

B1 28.8

B1

frictional force is 13.6N

A1

6

(c) at B, tension is $\frac{32\times1.2}{0.8}$

= 48N

B1

tension > friction
hence particle starts to move

E1

2

(d) when particle is next at rest,
work done against friction is EPE at B

13.6 × distance = 28.8

M1

distance is 2.1176

= 2.12 m

CAO

A1

2

(e) total distance is 5 + 2.1176

ft from M1 in (d)
or total distance × 13.6 = original EPE, 96.8

B1

1

= 7.12 m

total distance is 7.12 m

[14]

Q7.

(a) Initial EPE = $\frac{\lambda x^2}{2l}$

= $\frac{120 \times (0.5)^2}{2 \times 5}$

M1 for formula with extension 0.5

M1

= 3 J

A1

Initial KE is $\frac{1}{2} \times 0.4 \times 9^2 = 16.2$ J

When block is at A, $\frac{1}{2}mv^2 = 3 + 16.2$

M1

$v^2 = 19.2 \div 0.2 = 96$

Speed is 9.80 m s^{-1}

Accept $4\sqrt{6}$; *condone 9.79*

A1

4

(b) (i) Normal reaction is $mg = 0.4g$

M1

Frictional force is $0.4\mu g$ N

A1

Work done by frictional force is
5.5 × (0.4μg) or 2.2μg

m1

C of Energy, when at A, gives

19.2 – 5.5 × (0.4μg) = $\frac{1}{2}$ × 0.4 × v^2

Three terms, eg initial energy in (a) (=3 or 19.2); work done; KE at A.

M1

Fully correct

A1

19.2 – 2.2 μg = 0.2v^2
$v = \sqrt{96 - 11\mu g}$

Ft $v = \sqrt{(v^2\ in(a)) - 11\mu g}$

A1

6

(ii) Speed when rebounding is $\frac{1}{2}\sqrt{96 - 11\mu g}$

B1ft

Block is stationary at B

$\frac{1}{2}$× 0.4 × $\frac{1}{4}$(96 – 11μg) – 2.2μg

Three terms

M1

Two terms correct with sign

A1

= $\frac{120 \times (0.5)^2}{2 \times 5}$

Third term correct with sign

A1

$\frac{1}{2}$ × 0.1(96 – 11μg) – 2.2μg = 3
4.8 – 2.75μg = 3

Or 4.8 – 0.55 μg – 2.2μg = 3

A1

μ = 0.0668

A1

6

[16]

Q8.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,dx$

Condone lack of limits and 'dx'

M1

= $\left[\frac{\lambda x^2}{2l}\right]_0^e$

Must include limits from integral

A1

= $\frac{\lambda e^2}{2l}$

AG

A1

3

(b) (i) Using $T = \frac{\lambda x}{l}$, $7g = \frac{196x}{2}$

M1: could use 3g or 4g – at least 1 side correct

M1

$x = \frac{14g}{196}$

A1

= 0.7

A1

3

(ii) By C of Energy, when next at rest, EPE (initial) = PE change (for platform) + EPE (when at rest)

$$\frac{196 \times 0.7^2}{2 \times 2} = 4 \times g \times (0.7 - x) + \frac{196x^2}{2 \times 2}$$

M1: 3 terms (not including $\frac{1}{2}$ *mv^2)*

A1: 2 of 3 terms correct

M1A1

A1: all correct

A1

$2.45 = 2.8 - 4x + 5x^2$
$100x^2 - 80x + 7 = 0$

m1

$(10x - 7)(10x - 1) = 0$

A1

$x = 0.1$

[last A1: must give 0.1, not 0.1 and 0.7]

A1

6

Alternative

$$\frac{196 \times 0.7^2}{2 \times 2} = 4gX + \frac{196(0.7 - X)^2}{2 \times 2}$$

(M1)

(where X is distance moved upwards)

(A1)
(A1)

$4gX = 98 \times 0.7X + 49X^2$

(m1)

$X = 0,\ 0.6$

(A1A1)

(iii) Max speed when $T = mg$

M1

$4g = \dfrac{196x}{2}$

A1

$x = 0.4$

Or mid-point of values 0.2 and 0.6 above SC2

A1

3

[15]

Q9.

(a) $\text{EPE} = \dfrac{\lambda x^2}{2l}$

$= \dfrac{1800 \times (4)^2}{2 \times 6}$

$= 2400$ J

B1: Extension = 4.

M1: Substitution of 6, 1800 and their extension into EPE formula.

A1: Correct EPE

B1
M1
A1

3

(b) $\dfrac{1800 \times (x)^2}{2 \times 6} = \dfrac{1}{2} \times 200 \times 8^2$

$x^2 = 42.67$
$x = 6.53$ m

M1: Equation with EPE and KE terms, both correct.

A1: Correct extension. Accept $\dfrac{8\sqrt{6}}{3}$ *6.53 or AWRT 6.532.*

M1
A1

Distance from O is 12.5 m

A1: Correct distance. Accept 12.5 or AWRT 12.53.

A1

3

(c) Resistance force is 800 N
Work done by resistance force is $800 \times (x + 6)$

B1: Correct work done by resistance force.

B1

C of Energy gives

$\dfrac{1800 \times (x)^2}{2 \times 6} + 800 \times (x + 6) = \dfrac{1}{2} \times 200 \times 8^2$

M1: Three energy terms, KE, Work Done and EPE.

A1: EPE correct.

A1: Correct equation.

(M1A1)
(A1)

$150x^2 + 800(x + 6) = 6400$
$3x^2 + 16x - 32 = 0$
or $150x^2 + 800x - 1600 = 0$

A1: Correct quadratic equation with no brackets.

(A1)

$$x = \frac{-16 \pm \sqrt{16^2 + 4 \times 3 \times 32}}{2 \times 3}$$

dM1: Solving their quadratic equation with correct formula and correct substitution

dM1

$x = 1.5497$

A1: Correct positive solution stated.

Accept 1.54 or 1.55 or AWRT 1.55.

A1

Distance from O is 7.55 m

A1: Correct distance from O. Accept 7.55 or 7.54 or AWRT 7.55.

A1

OR
Use d for distance:
$800 \times d$

B1: Correct work done by resistance force.

(B1)

C of Energy gives

$$\frac{1800 \times (d - 6)^2}{2 \times 6} + 800 = \frac{1}{2} \times 200 \times 8^2$$

M1: Three energy terms, KE, Work Done and EPE.

A1: Seeing d – 6 in EPE

A1: EPE correct.

A1: Correct equation.

A1: Correct quadratic equation with no brackets.

(M1A1)
(A1A1)
(A1)

$150d^2 - 1000d - 1000 = 0$
$3d^2 - 20d - 20 = 0$

$$x = \frac{-20 \pm \sqrt{20^2 + 4 \times 3 \times 20}}{2 \times 3}$$

dM1: Solving their quadratic equation.

(dM1)

d = 7.55

A1: Correct distance from O. Accept 7.55 or 7.54 or AWRT 7.55.

(A1)

8

[14]

Q10.

(a) When $x \geq 22$, KE is $\frac{1}{2} \times 49 \times v^2$

EPE is $\frac{1078(x-22)^2}{2 \times 22}$

M1 for any $\frac{1078p^2}{2 \times 22}$

M1A1

Change in PE is $49 \times g \times x$

Conservation of energy:

$$\frac{1}{2} \times 49 \times v^2 + \frac{1078(x-22)^2}{2 \times 22} = 49 \times g \times x$$

M1 3 terms (KE, PE, EPE)
A1 2 terms correct
A1 all 3 terms correct

M1A1A1

$$\frac{49}{2}v^2 + \frac{49}{2}(x-22)^2 = 49gx$$

$$v^2 + (x-22)^2 = 19.6x$$

SC3 $\frac{49}{2}v^2 + \frac{49}{2}e^2 = 49g(e+22)$

[could use x for e]

$$5v^2 = 318x - 5x^2 - 2420$$

AG

A1

6

(b) If x is not greater than 22, cord is not stretched

B1

1

(c) At maximum value of x, $v = 0$

M1

$\therefore\ 5x^2 - 318x + 2420 = 0$

$$x = \frac{318 \pm \sqrt{318^2 - 4 \times 5 \times 2420}}{2 \times 5}$$

dep on M1 above

m1

$x = 54.76...$ or $8.84...$

A1 for either solution

A1

$= 54.8$

Needs to give a reason for deletion of second root. Both roots must be positive: one above 22, one below 22

E1

4

(d) (i) When speed is a maximum, $a = 0$

tension = gravitational force

or

$$\frac{d(5v^2)}{dx} = 318 - 10x$$

M1

$$\frac{1078(x-22)}{22} = 49g$$

= 0 at maximum speed $\Rightarrow$ *318 – 10x = 0*

A1

$x - 22 = 9.8$

$x = 31.8$

AG

A1

3

(ii) From part (a), $v^2 = 19.6 \times 31.8 - 9.8^2$

M1

$v = 22.96$

A1

Maximum speed is 23.0 m s^{-1}

2

[16]

Q11.

(a) When acceleration is zero, tension = gravitational force

$$\frac{784x}{16} = 80g$$

Both terms correct

M1

$x = 16$, $x + 16 = 32$ m

A1 for x = 16

A1

Length of cord is 32 m

A1

3

(b) (i) When bungee jumper comes to rest,

EPE = $\frac{784 \times x^2}{2 \times 16}$

M1

$$= \frac{49x^2}{2}$$

Change in PE = 80 × g × (16 + x)

Or 80 × g × 65 – (80g[16+ x])

(or 80g(49 – x))

M1

$$\frac{49x^2}{2} = 80 \times 9.8 \times (16 + x)$$

A1

$x^2 = 32x + 512$

$x^2 - 32x - 512 = 0$

AG

A1

4

(ii) $x = \frac{32 \pm \sqrt{32^2 + 2048}}{2}$

M1

x = 43.7128

A1

Distance below point of jump is 43.7 + 16 = 59.7 m
Distance between jumper and ground is 65 – 59.7

M1

= 5.29 m

Accept 5.287, 5.3

A1

4

[11]

Q12.

(a) EPE = $\frac{\lambda x^2}{2l}$

$= \frac{180 \times 0.8^2}{2 \times 1.2}$

M1

= 48 J

A1

2

(b) Using initial EPE = KE when string becomes slack:

M1

$48 = \frac{1}{2} \times 5 \times v^2$

A1F

$v = \sqrt{\frac{96}{5}}$

= 4.38 m s^{-1}

ft $\sqrt{\frac{'a'}{2.5}}$

A1F

3

(c) Normal reaction is 5g or 49

M1

Frictional force is $5g \times \mu$

m1 A1

Work done by frictional force is $5\mu g \times 2$

m1

$= 10\ \mu g$

A1

Stops at wall $\Rightarrow$ $10\mu g = 48$

m1 $10\ \mu g$ = 'a'

m1

$\mu = 0.490$

accept $\frac{24}{49}$ *OE*

A1

7

[12]

Q13.

(a) EPE is $\frac{\lambda x^2}{2l}$

$= \frac{300 \times 2^2}{2 \times 6}$

M1

= 100 J

A1

2

(b) Using $F = \mu R$:

Friction is $0.3 \times 4g$

M1

$= 1.2g$ or 11.76 N

A1

When string becomes slack:

Work done against friction = change in energy

m1

$1.2g \times 2 = 100 - \frac{1}{2}mv^2$

A1A1

$v^2 = 38.245$
Speed is 6.18 m s^{-1}
AG

A1
6
[8]

Q14.

(a) EPE = $\frac{\lambda x^2}{2l}$
$= \frac{300 \times (1.5)^2}{2 \times 4}$

M1

= 84.375
= 84.4 J

A1
2

(b) When string is slack, gain in PE is mgh
= 6 × g × 1.5 sin 30

M1

= 44.1 J

A1

KE = EPE – gain in PE

m1

= 84.375 – 44.1
= 40.275

A1

$\frac{1}{2}.6.v^2 = 40.275$
v = 3.66
AG

A1
5

(c) At A, PE gained above initial position is

$6 \times g \times 5.5 \sin 30$

Or PE above position string slack is 117.6

= 161.7 J

KE at A is – 77.3

B1

This is more than initial elastic potential energy

B1

$\therefore$ particle will not reach A

E1

Or

Using $v^2 = u^2 + 2as$

$a = 0.5g$ B1

$s = 1.37$ or 1.366 B1 [or 2.87 above starting point]

Hence stops before A E1

Vertical height above sling slack is 0.683

Vertical height above starting point is 1.435

3

[10]

Q15.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,dx$

M1

$= \left[\frac{\lambda x^2}{2l}\right]_0^e$

Needs limit of 0

A1

$= \frac{\lambda e^2}{2l}$

A1

Or

area under a straight line

= average force × distance

$= \frac{\lambda e^2}{2l}$

3

(b) (i) Using $T = \frac{\lambda x}{l}$

$5g = \frac{150 \times x}{0.6}$

M1

Extension is 0.196 m

A1

2

(ii) EPE = $\frac{\lambda x^2}{2l}$

$= \frac{150 \times (1.4)^2}{2 \times 0.6}$

M1

= 245 J

A1

2

(iii) When at point O,
EPE = 0
PE [relative to P] = 5 × g × 2

M1

KE [at O] = EPE [at P] – gain in PE [at O]

M1

$\frac{1}{2}mv^2 = 245 - 10g$

$\frac{1}{2}.5\,v^2 = 147$

A1

$v^2 = 58.8$ Speed is 7.67 m s^{-1}

A1

4

[11]

Q16.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,dx$

M1

$= \left[\dfrac{\lambda x^2}{2l}\right]_0^e$

Needs limit of 0

A1

$= \dfrac{\lambda e^2}{2l}$

AG

A1

Or

area under a straight line = average force × distance $= \dfrac{\lambda e^2}{2l}$

3

(b) (i) Using $T = \dfrac{\lambda x}{l}$

$5g = \dfrac{150 \times x}{0.6}$

M1

Extension is 0.196 m

A1

2

(ii) EPE = $\dfrac{\lambda x^2}{2l}$

$= \dfrac{150 \times (0.3)^2}{2 \times 0.6}$

M1

= 11.25 J

A1

2

(iii) When x above P,

EPE = $\dfrac{150 \times (0.3 - x)^2}{2 \times 0.6}$ =

for $\dfrac{150 \times (\ldots - x)^2}{2 \times 0.6}$

M1A1

PE [relative to P] = (–) 5 × g × x

for 5 × g × distance

M1

KE + EPE [at new point]
= EPE [at P] – gain in PE

4 terms, all signs correct, 2 terms correct

M1

$$\frac{1}{2}mv^2 + \frac{150 \times (0.3 - x)^2}{2 \times 0.6} = \frac{150 \times (0.3)^2}{2 \times 0.6} - 5gx$$

A1

$$\frac{1}{2}mv^2 + \frac{150 \times (x^2 - 0.6x)}{2 \times 0.6} = -5gx$$

Equation involving terms in v^2, x^2 and x only

m1

$$\frac{1}{2}.5.v^2 + 125x^2 - 75x = -49x$$

$v^2 = 10.4x - 50x^2$

A1

7

(iv) Particle is at rest when $v = 0$
$10.4x - 50x^2 = 0$

M1

$x = 0$ [not required]

Or $x = \frac{10.4}{50}$ = 0.208 m above P.

A1

2

[16]

Q17.

(a) $2g = \frac{49 \times x}{0.5}$

M1A1

<u>$x = 0.2$</u>

A1

3

(b) EPE = $\frac{49 \times (0.2)^2}{2 \times 0.5}$

M1

= 1.96 (J)

A1

2

(c) (i) $1.96 = \frac{49 \times x^2}{2 \times 0.5} + 0.8 \times 9.8 \times (0.2 + x)$

All terms attempted for M1

M1

-1 EE from A3

A3

$x^2 + 0.16x - 0.008 = 0$

A1

5

(ii) $x = \frac{0.16 \pm \sqrt{0.16^2 + 4 \times 0.008}}{2}$

M1

$x = 0.04$

$x = 0.04$ only identified

A1

2

[12]

Q18.

(a) KE is loss in PE
= 4 × g × 1.5

M1 for mgh = 58.8 and then find v without finding KE

M1

= 6g J or 58.8 J

A1

2

(b) When 3.5 m below O, extension is 2 m

EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(2)^2}{2 \times 1.5} = \frac{4}{3}\lambda$

M1

Change in potential energy of the particle is
$4 \times g \times 3.5$

M1

$= 14g$ or 137.2

A1

$\frac{4}{3}\lambda = 14g$
$\lambda = 102.9$ N or 103 N
AG

A1

4

(c) When particle is 2.7 m below O,
EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(1.2)^2}{2 \times 1.5} = 49.392$
Accept 49.44 [from 103]

M1A1

Change in potential energy of the particle
[from initial position] is
$4 \times g \times 2.7 = 10.8g$ or 105.84

B1

Conservation of energy:

$105.84 = \frac{1}{2} \times m \times v^2 + 49.392$
M1 for 3 terms and $4 \times g \times h$

M1

$2v^2 = 56.448$
Speed is $5.3126 \text{ m s}^{-1} = 5.31 \text{ m s}^{-1}$
CAO

A1

5

[11]

Q19.

(a) EPE is $\frac{\lambda x^2}{2l}$

$= \dfrac{200(0.5)^2}{2 \times 2}$

M1

= 12.5 J

A1

2

(b) When string becomes slack,

using $\frac{1}{2}mv^2$ = loss in EPE

NB Using $\sqrt{5}$ to answer (a) and thus (b) $\Rightarrow$ no marks

M1

$\frac{1}{2} \times 5 \times v^2 = 12.5$

A1

Speed is $\sqrt{5}$ m s^{-1}

AG

A1

3

(c) Resolving vertically, $R = 5g$

B1

$F = \mu R$

M1

$0.4 \times 5g = 2g$

M1

Using change in energy = work done:

$2g \times 0.5 =$

M1 for force × distance

M1

$\frac{1}{2} \times 5 \times \left(\sqrt{5}^2\right) - \frac{1}{2} \times 5 \times v^2$

A1 first term (or 12.5)

A1 second term (inc –)

A1,A1

$$9.8 = 12.5 - \frac{5}{2}v^2$$

$v^2 = 1.08$
Speed is 1.04 m s^{-1}

A1

7

[12]

Q20.

(a) $\frac{100}{0.4}e = 10 \times 9.8$

Use of Hookes law and equilibrium

M1

$e = 0.392$ m

Correct length

A1

2

(b) $EPE = \frac{1}{2} \times \frac{100}{0.4} \times 0.6^2 = 45$ J

Use of EPE formula

M1

AG

Correct value from correct working

A1

2

(c) (i) $45 = \frac{1}{2} \times \frac{100}{0.4}(x - 0.4)^2 + \frac{1}{2} \times 10v^2 + 10 \times 9.8(1 - x)$

Expression for EPE with $(x \pm 0.4)^2$

M1

Correct EPE

A1

Four term energy equation

M1

$45 = 125(x - 0.4)^2 + 5v^2 + 98(1 - x)$

Correct GPE

B1

Correct equation

A1

$5v^2 = 98x - 98 + 45 - 125x^2 + 100x - 20$

Solving for v^2

dM1

$v^2 = 39.6x - 25x^2 - 14.6$ AG

Correct result from correct working

A1

7

(ii) $39.6x - 25x^2 - 14.6 = 0$

$25x^2 - 39.6x + 14.6 = 0$

$x = \dfrac{39.6 \pm \sqrt{39.6^2 - 4 \times 25 \times 14.6}}{2 \times 25}$

Solving quadratic

M1

= l or 0.584

Correct solutions

A1

$x = 0.584$

Appropriate value selected

SC Only correct answers given, award M1A1.

A1

3

[14]

Q21.

(a) $\text{EPE} = \dfrac{1}{2} \times \dfrac{30}{0.5} \times 1.3^2 = 50.7\text{ J}$

use of EPE formula

M1

correct EPE

A1

2

(b) (i) $50.7 = \dfrac{1}{2} \times 2v + \dfrac{1}{2} \times \dfrac{30}{0.5} \times 0.8^2$

three term energy equation

two terms correct

M1

all terms correct

A1

$50.7 = v^2 + 19.2$ AG

A1

solving for v

dM1

$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$

correct v from correct working

A1

5

(ii) $50.7 = \frac{1}{2} \times 2v^2$

M1

two term energy equation

correct equation

A1

$v = \sqrt{50.7} = 7.12 \text{ ms}^{-1}$

correct velocity

A1

3

(c) $\frac{1}{2} \times 2v^2 = 50.7 - 1.8 \times 0.1 \times 2 \times 9.8$

finding friction force

M1

correct friction force

A1

three term energy equation

M1

correct equation

A1

$v = \sqrt{47.172} = 6.87 \text{ ms}^{-1}$

correct velocity

A1

5

[15]

Q22.

(a) $12.5 = \lambda \times \frac{0.1}{0.4}$

M1: subs. A1:All correct

M1A1

$\lambda = 50$

A1

3

(b) $\text{EPE} = \frac{50 \times (0.1)^2}{2 \times 0.4}$

Subs.

M1

$= 0.625$ J

PI A1: all correct

A1

$0.625 = \frac{1}{2} \times 0.2 \times v^2$

M1 use of principle

M1

ft EPE

A1F

$v = 2.5$ ms^{-1}

ft EPE

A1F

5

[8]

Q23.

(a) KE = Initial PE

Alternative for (a):

$\frac{1}{2}(50)\,v^2 = (50)\,g\,(20)$

Use of $v^2 = u^2 + 2as$

M1

$\therefore v^2 = 40g$

$v^2 = 0^2 + 2g\,(20)$

$v \approx 19.8\ \text{ms}^{-1}$

$v = 19.8\ \text{ms}^{-1}$

A1

2

(b) (i) EPE after searching = PE at start

M1

= 50g (32)
= 15 680 J

Alternative for (b)(i):
EPE after stretching = PE + KE at natural length

$= 50\,(g)\,(12) + \frac{1}{2}(50)\,(19.8..)^2$

AG = 15680 J

A1

2

(ii) $\frac{1}{2}k\,(32 - 20)^2 = 15\,680$

B1 for $\frac{1}{2}k\,x^2$; M1 for equation

M1B1

$72k = 15\,680$

$k = 218$

A1 CAO

A1

3

[7]

Q24.

(a) EPE = $\frac{1}{2}(50)(0.03)^2$

Use of correct formula

M1

= 0.0225 J

A1

2

(b) KE + PE = EPE

$\frac{1}{2}mv^2 + mgh = \text{EPE}$

KE or PE term correct

B1

$\frac{1}{2}(0.02)v^2 + (0.02)(9.8)(0.03) = 0.0225$

Equation formed

Equation correct; ft EPE

M1 A1f.t.

$v \approx 1.29\ \text{ms}^{-1}$

AG

A1

4

(c) Energy remains constant –
No KE at end, no EPE at end
$\therefore\ (0.02)(9.8)h = 0.0225$

M1A1

$\therefore\ h = 0.11$ m

ft EPE *2 sig fig accuracy*

A1f.t.

3

i.e. 11 cm

Alternatives to part (c)

Energy remains constant
KE at point of release = PE on reaching max height
$\frac{1}{2}(0.02)(1.29)^2 = (0.02)gh$

(M1)

$\therefore\ h = 0.849...$

(A1)

$\therefore$ height above initial position = 0.849... + 3 = 0.11m

ft their h value + 3

(A1f.t.)

or Use of $v^2 - u^2 = 2as$

(M1)(A1)

$s = 0.849$ distance = 0.849 + 3

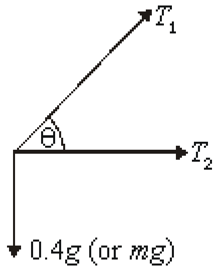
ft their (s + 3) total

(A1f.t.)

[9]

Q25.

(a)



Three forces evident;
T_1, T_2 clear – on diagram or in calculations

B1

1

(b) Vertically $T_1 \sin \theta = 0.4g$

Resolve vertically – component evident
$T \sin \theta = mg$ seen

M1 A1

$T_1 \left(\frac{4}{5}\right) = 0.4g$

M1

$T_1 = 0.5g = 4.9$N

AG obtained

A1

3

(c) $T_2 = 0.1k$

B1

1

(d) $a = r\omega^2 = 0.3 \times 5^2$

Calculation of a seen

B1

Force = $T_2 + T_1 \cos \theta$

(Both terms of horizontal resultant force attempted)

M1

$= 0.1k + 4.9 \times \frac{3}{5}$

Previous expression / result substituted

A1

$F = ma$

Their 'a' and 'F' (Dependent on first M1)

m1

$k = 0.6$

AG

A1

5

(e) EPE = $\frac{kx^2}{2} = \frac{0.6(0.1)^2}{2}$

$\frac{kx^2}{2}$ *seen and attempt to use*

M1

= 0.003 J

A1

2

[12]

Q26.

(a) Rope taut (momentarily) after 14m,

so PE = KE

$70g(14) = \frac{1}{2}(70)v^2$

attempt at conservation of energy
or $v^2 = u^2 + 2as$

M1

$v^2 = 28g$

$v \approx 16.6$

AWRT

A1

2

(b) (i) Initial PE = EPE at end

$70g(14 + x) = \frac{1}{2}(196)x^2$

attempt at conservation of energy equation

fully correct
For EPE (any stage)

M1 A1 B1

$70(14 + x) = 10x^2$

$7(14 + x) = x^2$

$0 = x^2 – 7x – 98$
Correct quadratic – AG

A1
4

(ii) $0 = (x – 14)(x + 7)$
any valid method for solving quadratic

M1

$x = 14$ or $–7$

$\therefore\ x = 14$ m

A1
2

(iii) Hooke's law: $T = kx$

$\therefore\ T = 196(14)$
Hooke's Law used; ft their x

B1f.t.

Newton's law: $70g – T = 70a$

M1 attempt F = ma

M1A1

$\therefore\ \dfrac{70(9.8) - 196(14)}{70} = a$

$–29.4\ ms^{–2}$
accept + or –

A1
4

[12]

(a) Rope taut (momentarily) after 14m,

so PE = KE

$70g(14) = \dfrac{1}{2}(70)v^2$

attempt at conservation of energy
or $v^2 = u^2 + 2as$

M1

$v^2 = 28g$

$v \approx 16.6$
AWRT

A1
2

(b) (i) Initial PE = EPE at end

$70g(14 + x) = \frac{1}{2}(196)x^2$
attempt at conservation of energy equation
fully correct
For EPE (any stage)

M1 A1 B1

$70(14 + x) = 10x^2$

$7(14 + x) = x^2$

$0 = x^2 - 7x - 98$
Correct quadratic – AG

A1
4

(ii) $0 = (x - 14)(x + 7)$
any valid method for solving quadratic

M1

$x = 14$ or -7

$\therefore\ x = 14$ m

A1
2

(iii) Hooke's law: $T = kx$

$\therefore\ T = 196(14)$
Hooke's Law used; ft their x

B1f.t.

Newton's law: $70g - T = 70a$
M1 attempt F = ma

M1A1

$$\therefore \frac{70(9.8) - 196(14)}{70} = a$$

$-29.4\ ms^{-2}$

accept + or –

A1

4

[12]

Q27.

(a) EPE = $\frac{1}{2} \times \frac{40}{2} \times 3^2$ = 90 J

Finding EPE

M1

***ag** Correct EPE from correct working*

A1

2

(b) 90 = $\frac{1}{2} \times 5v^2$

Use of EPE = KE

M1

Correct equation

A1

$v^2 = 36$
$v = 6$

***ag** Correct speed from correct working*

A1

3

(c) $EPE = \frac{1}{2} \times \frac{40}{2} \times 1^2 = 10$ J

Finding EPE 3 metres from O

M1

Correct EPE

A1

$90 - 10 = \frac{1}{2} \times 5v^2$

Using EPE lost = KE

M1

Correct equation

A1

$v^2 = 32$
$v = 5.66 \text{ ms}^{-1}$ (to 3 sf)

Correct speed

A1

5

[10]

Q28.

(a) $$\text{EPE} = \frac{30 \times 0.8^2}{2 \times 2} = 4.8\text{ J}$$

Use of EPE formula with 0.8

M1

Correct EPE

A1

2

(b) (i) $4.8 = 0.15 \times 9.8 \times 2.8 + \frac{1}{2} \times 0.15 \times v^2$

Three term energy equation

Accept $0.684 = \frac{1}{2} 0.15 v^2$

M1

$$v = \sqrt{\frac{4.8 - 4.116}{0.075}} = 3.02\text{ ms}^{-1}$$

Correct equation

A1

Solving for v

m1

ag *Correct v from correct working*

A1

4

(ii) $4.8 = 0.15 \times 9.8 \times 2.8 + 0.15 \times 9.8h$

Three term energy equation using height above O
Accept 0.684 = mgh

M1

Correct equation

A1

$$h = \frac{4.8 - 4.116}{1.47} = 0.465 \text{ m}$$

Correct height above O
Accept 0.47 or 0.46

A1

As 0.465 < 2 the string does not become taut

Correct conclusion
Alternative
M1: Use of constant acceleration equation
A1: Correct equation
A1: Correct height
A1: Correct conclusion

A1

4

[10]

Q29.

(a) $20 \times 9.8 = \frac{0.7\lambda}{2}$

Use of T = mg

M1 A1

$$\lambda = \frac{2 \times 20 \times 9.8}{0.7} = 560$$

Correct result from working

A1

3

(b) (i) $20 \times 9.8L = \frac{560(L-2)^2}{2 \times 2}$

Two term energy equation
Correct terms
Correct signs

M1 A1 A1

$196L = 140L^2 - 560L + 560$

Expanding and simplifying

m1

$5L^2 - 27L + 20 = 0$

Correct result from correct working

A1

5

(ii) $L = \dfrac{27 \pm \sqrt{27^2 - 4 \times 5 \times 20}}{2 \times 5}$

Solving a quadratic

M1

= 4.51 or 0.886

Correct solutions

A1

L = 4.51

Selecting the appropriate solution

A1

3

[11]

Q30.

(a) (i) KE = 2 × 9.8 × 4 = 78.4 J

Use of KE = change in PE with h = 4

M1

Correct energy

A1

2

(ii) $78.4 = \frac{1}{2} \times 2 \times v^2$

Use of kinetic energy or constant acceleration formula to form an equation in v based on a fall of 4 metres

M1

Correct equation

A1

$v = \sqrt{78.4}$ = 8.85 ms^{-1}

Correct v

A1

3

(b) (i) $78.4 + 19.6x = \frac{80}{2 \times 4} x^2$

Calculation of EPE shown

M1

Correct EPE

A1

Three term energy equation

M1

$0 = 10x^2 - 19.6x - 78.4$

ag *Correct equation from correct working*

A1

4

(ii) $x = \dfrac{19.6 \pm \sqrt{19.6^2 - 4 \times 10 \times (-78.4)}}{2 \times 10}$

Solving the quadratic equation

M1

= 3.95 or – 1.99

Correct solutions

A1

Max L = 7.95 m

ft Adding 4 to their x

A1ft

3

(c) No air resistance

Appropriate assumption

B1

1

[13]

Q31.

(a) EPE = $\dfrac{12 \times 1^2}{2 \times 0.5} = 12\text{ J}$

Calculating EPE

M1

Correct EPE

A1

2

(b) (i) KE = 12 – 0.5 × 9.8 × 1

Using PE to obtain KE

M1

= 7.1J

Correct answer from correct working

A1

2

(ii) KE = 7.1 – 0.5 × 9.8 × 1

Using PE to obtain KE

M1

= 2.2J

Correct answer

A1

2

(iii) $0.5 \times 9.8x + \frac{12x^2}{2 \times 0.5} = 2.2$

Correct EPE

B1

Three term energy equation

M1

Correct equation

A1

$12x^2 + 4.9x - 2.2 = 0$

$x = \frac{-4.9 \pm \sqrt{4.9^2 - 4 \times 12 \times (-2.2)}}{2 \times 12}$

Solving quadratic

m1

$x = 0.270$ or -0.679

Correct solutions

A1

Max height = 0.5 + 0.270 = 0.770m

Calculating actual height

m1

Correct height

A1

7

[13]

Examiner reports

Q1.

In this question, students were able to demonstrate their understanding of principles of energy shown by a variety of correct approaches in part (b). The best approaches had clearly labelled diagrams which helped students obtain correct and consistent expressions for different types of energies. The mean mark for part (a) was 51%, the mean mark for (b) was 43% and the mean mark for (c) was 19%.

Part (a) was meant to be a simple application of conservation of energy when the cord reached its natural length. Many students applied the formulae correctly but did not give the final answer to two significant figures, thus losing the final mark. Required accuracy should match the value of g given in the question. A less common error was to use the height of the bridge above the river rather than the 25 m required.

There were many different approaches to part (b) and if students had used diagrams they would have been less confused and obtained more marks. The approaches seen were:

- calculation of the initial total energy (36750 J) and to compare this with the energy required for Dominic to reach the river (40000 J), deducing that there was insufficient energy to do so
- forming an energy equation and attempting to find the speed of Dominic at the surface of the river, deducing that he did not reach the river as v^2 was negative
- using x as the maximum extension, forming an energy equation to deduce that x = 23.6 m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the maximum length, forming an energy equation to deduce that x = 48.6 m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the minimum distance above the river, forming an energy equation to deduce that x = 1.4 m deducing that Dominic would not get wet.

All these methods proved to be successful for some students. However, without clear diagrams other students often got confused about what they were finding, leading to incorrect combinations of the different types of energies. Sometimes an x that had started out as a maximum length became an extension instead.

Part (c) linked to part (b) and required students to realise that if Dominic was not a particle then he would have height and hence could get wet if he was taller than 1.4 m. Any reference to air resistance here scored zero marks.

Q4.

Many students were unable to show convincingly that the work done in stretching the string was $\frac{\lambda e^2}{2l}$. An integration using a variable, usually x, was required. A number of students were penalised for using either $\int_0^e \frac{\lambda e}{l}\, de$ or $\int \frac{\lambda x}{l}\, dx$ (without including limits). Parts (b)(i) and (b)(ii) were usually answered correctly although a minority of students used the incorrect formulae for tension and elastic potential energy. In part (b)(iii), most students included terms in kinetic energy, gravitational potential energy and elastic potential energy but many did not include elastic potential energy at both A and the point when its speed was 0.8 ms^{-1}. Another common error was in using the same variable for the height that the particle moved (in the potential energy gained) and for the extension when the particle's speed was 0.8 ms^{-1}.

Q5.

This question was designed to discriminate between grades A and A* and it was common for students who scored very well in the rest of the paper to find this question challenging. Many students were successful in part (a) but some students considered the initial Elastic Potential Energy in the string rather than the initial tension. Part (b) was also completed well by many students although a number forgot the square in their elastic potential energy term. To solve part (c) of this question, students needed to consider four terms, which were the initial elastic potential energy, the work done by friction, the change in potential energy of particle B and the final elastic potential energy. Many students forgot that the string had a non-zero elastic potential energy when particle B was at rest. Relatively few students considered the work done by the frictional force which was the friction acting on particle A (31.36N) multiplied by the distance A had moved (0.46m). A few students incorrectly used $v^2 = u^2 + 2as$, ignoring the fact that constant acceleration formulae cannot be used with elastic strings.

Q6.

Part (a) of this question was answered successfully by most candidates. Although many candidates were successful in part (b), a few attempted to use kinetic energy. Relatively few candidates were successful in part (c), with most candidates stating that when the particle was at *B* the string was stretched and hence the string had elastic potential energy and thus the particle must move. The fact that the particle would only move again if the tension in the string was greater than the friction, found in part (b), was only considered by the better candidates. Those candidates who had answered part (b) correctly usually answered parts (d) and (e) correctly.

Q7.

In part (a) most candidates found the initial EPE correctly, but a number forgot to add on the initial KE to find the energy of the block at *A*. In part (b)(i), most candidates found the frictional force to be 0.4 μg but often failed to multiply this by 5.5 to obtain the work done by the frictional force. Other candidates used $v^2 = u^2 + 2as$, ignoring the fact that constant acceleration formulae cannot be used with elastic strings. In part (b)(ii), candidates often used the speed after rebounding to be half the speed of the block before the impact, but when finding the kinetic energy forgot to square the half, when calculating v^2. The required equation on part (ii) included three terms, KE when leaving B, work done by friction and the EPE in the string when the block came to rest at B. Unfortunately many candidates forgot that the string had a non-zero EPE when the block was at B.

Q8.

Many candidates were unable to show convincingly that the work done in stretching the string

was $\frac{\lambda e^2}{2l}$ An integration using a variable, usually x, was required.

Part (b)(i) was usually answered correctly. In part (b)(ii), most candidates appreciated that they needed to consider gravitational potential energy and elastic potential energy. However, many candidates did not appreciate that the string was still stretched when the 4 kg block was next at rest. Often candidates used terms in kinetic energy, gravitational potential energy and elastic potential energy and then equated the speed to zero.
In part (b)(iii), the extension x when the speed is at a maximum was usually found by using $T = mg$. A few candidates used techniques involving the maximising of v^2 ; impressively, sometimes these were totally successful.

Q9.

Part (a) was answered well by most candidates. Most candidates answered part (b) correctly, although a few tried to use tension rather than elastic potential energy. In part (c), most candidates appreciated that they needed to consider kinetic energy, work done and elastic potential energy, although a few also tried to include gravitational potential energy. Many candidates did appreciate that the work done was force × distance but did not realise that this distance was not the extension of the string.

Q10.

Most candidates appreciated that they needed to consider kinetic energy, gravitational potential energy and elastic potential energy. However the extension in the cord was not equal to the height dropped by the bungee jumper and this caused difficulty to many candidates.

In part (b) most candidates gave a reasonable explanation why x had to be greater than 22.

In part (c) most candidates knew that $v = 0$ when x had a maximum value, but only the better candidates appreciated that, since x had to be greater than 22, x had to be the larger of their two solutions. This statement was required for full credit to be given.

In part (d) (i) the extension x when the speed is at a maximum was found either by maximising $5v^2$ or by using $T = mg$, and candidates used these two approaches in equal numbers with equal success. Part (d) (ii) was answered well.

Q11.

Many candidates experienced difficulty with this question. In part (a), candidates rarely used the fact that the acceleration of the bungee jumper was zero when the tension in the cord was equal to the gravitational force. Candidates tried to use energy considerations in part (b)(i), but did not use the correct difference in height for potential energy. There was a significant amount of 'creative' algebra to obtain the printed equation. In part (b)(ii), most candidates solved the given equation quickly, and many then subtracted this from 65 to find the distance between the bungee jumper and the ground. The three significant figure solution to the quadratic resulted in a two significant figure answer which was not penalised. This rarely concerned candidates; only the better ones found the four significant figure solution to the quadratic in order to obtain a three significant figure answer.

Q12.

Parts (a) and (b) of this question were answered well. In part (c), many complicated methods were used to find the work done; often speed was considered and it was common for the work done, exactly 48 J from part (a), to become a value fairly close to 48 J. Most candidates found the frictional force to be $5g\mu$, but relatively few candidates used work done by friction to be $5g\mu$ multiplied by the distance moved, which was 2 m.

Q13.

Candidates found this question straightforward and answered it well

Q14.

Most candidates answered part (a) well. In part (b), most candidates attempted to find the final kinetic energy. Unfortunately, a few candidates found an expression for the final speed without showing the result of the intermediate calculations, and merely wrote the result as given in the question, 3.66 m s^{-1}. Such candidates were penalised. Candidates should appreciate that a question requiring them to "Show that …" means they cannot simply write down the answer. In part (c), candidates used a variety of methods, generally based on energy, to show that the particle could not reach A.

Q15.

Part (a) tested work done = $\int F \, dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and x = e; or work done = maximum force × half the distance moved, without a clear justification.

Part (b)(i) and (ii) were completed well. In part (b)(iii), most candidates attempted to conserve kinetic energy, potential energy and elastic potential energy, but often the signs of the three terms in the equation $\frac{1}{2}mv^2 - 245 + 10g = 0$ were incorrect.

Q16.

Part (a) tested work done = $\int F \, dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and x = e; or work done = maximum force × half the distance moved, without a clear justification.

Part (b)(i) was completed well, as was part (b)(ii), although a number of candidates did write down $\frac{150 \times 0.9^2}{2 \times 1.2} = 11.25$, which evidently it is not. In part (b)(iii), many tried to use the conservation of kinetic energy, potential energy and elastic potential energy. The printed result enabled many of those who obtained an incorrect answer to correct their working. Commonly, this was because they had used $\frac{\lambda(0.9 - x)^2}{2l}$ instead of $\frac{\lambda(0.3 - x)^2}{2l}$ for the elastic potential energy. Part (c) was completed well, but some candidates did not explicitly exclude $x = 0$.

Q17.

Candidates were often very successful in parts (a) and (b). Most knew how to start part (c)(i) but elastic energy terms were sometimes omitted and there were many errors in potential energy terms. Part (c)(ii) was successful but not all selected the appropriate solution.

Q18.

In part (a), many candidates used conservation of energy, not to find the kinetic energy when the string became taut, but to find the velocity when the string became taut. Then they used KE = $\frac{1}{2}mv^2$ to find the kinetic energy when the string became taut. The simpler method of using conservation of energy with gain in kinetic energy equalling the loss in potential energy was rarely seen; this of course immediately gave the required kinetic energy. In part (b), most candidates used the change in potential energy to be the change in EPE and readily found λ.

Q19.

Parts (a) and (b) caused few problems for the vast majority of candidates. In part (c) it was common to see the frictional force to be $5g \times 0.4 = 19.6$ N, but this was often not multiplied by the distance moved, 0.5 (m), to find the work done.

Q20.

Parts (a) and (b) of this question were done very well, with many candidates gaining full marks on both parts. Part (c) (ii) was also done well, but some candidates did not show both solutions to the quadratic as part of their working. In questions like this candidates should be encouraged to show both solutions and then select the appropriate solution in the context under consideration. Part (c)(i) was very challenging and there were relatively few correct solutions. The three main problems were:

- Not including the elastic potential energy when released,
- Using an incorrect expression for the extension of the string when forming an expression for the elastic potential energy,
- Using an incorrect distance when forming an expression for the gravitational potential energy.

Q21.

Part (a) was done very well, with very few incorrect responses. Part (b) of the question was also done fairly well, with the printed answer helping some candidates. The main issue here was that some candidates did not use the correct distances to find the elastic potential energies. Part (c) was much more demanding. Many candidates were able to calculate the magnitude of the friction force correctly, but far fewer were able to incorporate this correctly into an energy equation.

Q22.

Part (a) was done well, with candidates able to obtain the printed result convincingly. In part (b) there were many successful solutions. However, some candidates assumed the force on the particle to be constant and applied Newton's Second Law followed by the equations of constant acceleration.

Q23.

There was an excellent response to this question. The formula for elastic potential energy now seems well known. The printed answer in part (b) helped candidates correct their work when they obtained a mismatch. Only a few candidates failed to use the correct extension in order to find k.

Q24.

There was a very good response to this question. The formula for elastic potential energy seemed to be well known and units were almost totally consistent and correct. The printed answer in part (b) helped candidates to correct their work when they obtained a mismatch. In part (c) a mark was often lost for failing to add on the initial 3 cm, if an approach using kinetic and potential energy was used.

Q25.

This proved to be a demanding question although a significant proportion of candidates solved it well. The force diagram in part (a) was rarely correct with equal tensions shown or, even worse, the tension in OP missing and sometimes a normal reaction appearing at P. Those candidates who had not clearly marked different tensions were able to recover the mark if their working out indicated the use of two different tensions.

In parts (b) and (d) many candidates struggled to set up the equations required. The answer in part (b) was arrived at through incorrect methods to match the answer given. In part (d) mrw^2 was often seen, but a number of candidates thought that this was directed along QP or outwards from P.

Parts (c) and (e) were more successful and it was pleasing to see the correct formulae in use.

Q26.

This was the most discriminating question on the paper. Part (a) did not particularly trouble any candidate.

In part (b), candidates who had been taught the $\frac{\lambda x^2}{2l}$ and the $\frac{\lambda x}{l}$ formulae were often not able to complete the question successfully. The specification is clear in its expectation. Weaker candidates often got muddled by mixing up energy terms from different points in the motion. It was pleasing to see that where a candidate had drawn clear diagrams the answer followed easily. It is worrying that many did not score two marks on the quadratic equation. Incomplete methods, invalid methods and failing to clearly rule out –7 as a solution were some of the reasons. Part (b)(iii) was scarcely seen correct, as many candidates did not realise that $mg - T = ma$ was required.

Q27.

This question was answered well by the majority of candidates, who were probably helped by the printed answers.

Part (c) caused a few more difficulties for some candidates than the earlier parts. The main errors were to assume that the kinetic energy would be 10 J or to include gravitational potential energy in some way.

Q28.

Part (a) of this question was generally done very well by the candidates. Part (b) was found to be more challenging. A few candidates did not attempt either parts (b)(i) or (b)(ii), with others having a go at one but not the other. In both parts, some candidates tried to use constant acceleration equations instead of energy methods. The candidates tended to be more successful with part (b)(i) than with part (b)(ii). In part (b)(ii), quite a few candidates formed and solved an energy equation, but did not interpret their result.

Q29.

Part (a) was done well, with the printed answer helping some candidates. However, some candidates did try to use elastic potential energy in their solutions. In part (b), several candidates struggled to form a correct equation. Those who did normally went on to produce the required equation. Common errors were to omit the gravitational potential energy or to use $L - 2$ when finding the gravitational potential energy. Part (c) was often done well, but there were some candidates who seemed unaware of any of the techniques that could be used to solve a quadratic equation.

Q30.

There were many good attempts at this question. Part (a) was generally done well, with some candidates basing their approach on the use of constant acceleration equations. A few candidates used an incorrect distance and made very little useful progress with the question. In part (b), several candidates seemed to have been helped by the printed result. Some candidates did not justify all the steps that they took. While most candidates were able to solve the quadratic equation without difficulty, it did cause problems for some. Some candidates did not add the natural length to the extension to find the maximum length. Many candidates gave good assumptions, but some quoted assumptions that would be difficult to describe as important, or ones that had been covered in the statement of the question.

Q31.

The candidates found the last part of this question very challenging, but the earlier parts straightforward. In part (a), a few candidates omitted the factor of 1/2 from the formula for the elastic potential energy. Very few were unable to find the kinetic energy given in part (b)(i). In part (b)(ii), a few candidates made errors by using the distance in their energy calculations, but the majority of candidates produced the correct value. Most of the candidates made very little progress with part of the question. In (b) (iii) the major stumbling block was not to include the elastic potential energy in their equations. Those who did include the elastic potential energy could make some progress with the solution. There were only a few correct answers to this part of the question.